



SCS1306 – Linear Algebra
Octave Practical Sheet - 01

Check your Operating System type and version.

Go to the website: <https://octave.org/download>, which contains Octave installation files, and download the Octave version suitable for your Operating System type and version.

Install Octave on your computer by double-clicking the Octave installation file.

Upon installing Octave, you will notice that it installs 2 icons on your desktop.

- One is the CLI (Command Line Interface).
- The other is the GUI (Graphical User Interface).

There are 2 methods for running commands in Octave.

- One method is to run commands in an interactive mode.
- The other method is to type commands in a .m file and to run that file.

Do the following exercises:-

1. Open the Octave CLI and type the following at the prompt and check the result for each.

Arithmetic Operators	Symbols
Addition	$a + b$
Subtraction	$a - b$
Multiplication	$a * b$
Exponentiation	$a ^ b$ or $a ** b$
Division	a / b

- $7 + 3$
- $89 - 9$
- $10 - 20$
- $4 * 7$
- $88 / 22$
- $3 / 7$
- $2 ^ 2$
- $2 * 3 + 1$
- $2 * (3 + 1)$

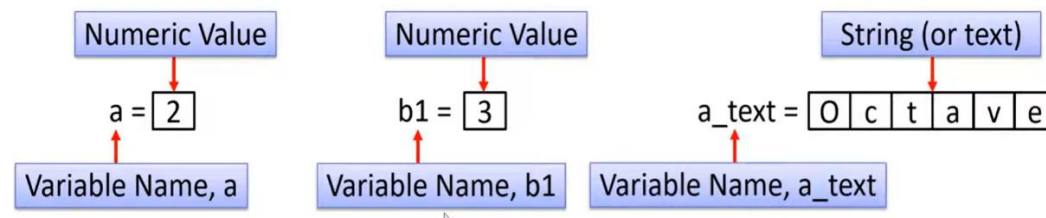
2. Evaluate the following using the Octave CLI

- $9^2 + 7^3 * (3 + 8 - 9/3) * 2^3$
- $8^2 - 7 * 2 * (8 + 6) / 2$
- $2 + 3 - 5 * 6 / (7 - 5)^5$
- $3 * 9 / 3 - 7^4 * (7 - 3)$

3. Evaluate the following using the Octave CLI

- $7 - 66 + 4 * \cos(0)$
- $\cos(0) * 33 / 3 - 67 + 8$
- $\cos(0) + \cos(0) + \cos(0)$
- $5600^{\cos(0)}$
- $55 + 7 / \cos(0)$
- $3 / \sin(0)$
- $6.789789789 / \sin(0)$

4. Practice variables



`[variablename] = [expression] <enter>`

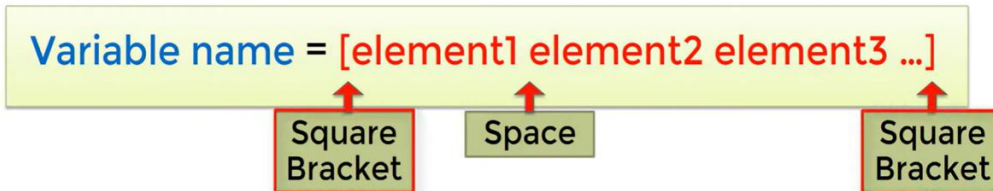
- `a=2`
- `a=3;`
- `b`
- `b=(1+3)/2`
- `d=b+1`
- `a=a+1`

- `who`
- `whos`
- `clear a b`
- `clear all`

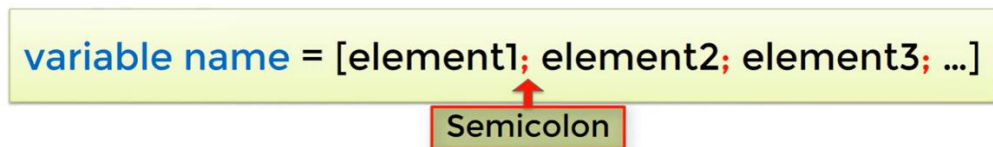
- `a!bc=2`
- `a bc=2`
- `1abc=2`
- `abc=2`
- `Abc=2`
- `a_text='octave'`

5. Practice Vectors

$$a = [3 \quad 5 \quad -2 \quad 0 \quad -1]$$



$$g = \begin{bmatrix} 3 \\ 5 \\ -2 \\ 0 \\ -1 \end{bmatrix}$$



- `a=[3 5 -2 0 -1]`
- `a(3)`
- `a(5)`
- `a(3)=2`
- `a(6)=-1`
- `a(10)=-2`
- `a([1 4])`
- `a([1 4]) = [6 -3]`
- `b=1:5`
- `b(2:4)`
- `c=0:0.1:0.5`
- `d=5:-1:1`
- `g=[3;5;-2;0;-1]`
- `h=(1:5)'`
- `x=1:5`
- `x'`

6. Practice Matrices

$$A = \begin{bmatrix} 3 & 5 & -1 \\ 7 & 8 & 5 \\ 0 & -1 & 2 \end{bmatrix}$$

variable name = [element1 element2 element3; ...]

Space

Semicolon

What are the outputs of the following commands?

- `A = [3 5 -1; 7 8 5; 0 -1 2]`
- `A(2,3)`
- `A([1 3],3)`
- `A(1:2,3)`
- `A(:,3)`
- `I=eye(3)`
- `B=ones(3)`
- `B=ones(3,5)`
- `C=zeros(3)`
- `C=zeros(3,5)`
- `X=1:12`
- `E=reshape(X,3,4)`
- `H=[B C]`
- `J=[B;C]`
- `[row col]=size(H)`

7. Matrix Operations

Matrix Operations	Octave/Matlab Syntax
Addition of Matrices A and B	A + B
Subtraction of Matrix B from A	A - B

Note: Size of matrices **A** and **B** must be the same to add and subtract

Matrix Operations	Octave/Matlab Syntax
Scalar, λ , Multiplication on Matrix A	$\lambda * A$
Matrix Multiplication of Matrices A and B	$A * B$

Note: Matrix multiplication requires multiplication conformability

What is the output of the following operation? $a + b$, ($c = a + b$)

- If the values of the Matrices are: -
- $a = [9 \ 3 \ 4; 5 \ 8 \ 7]$
- $b = [7 \ 4 \ 3; 8 \ 7 \ 2]$

8. What is the output of the following operation? $a - b$

- If the values of the Matrices are:-
- $a = [2 \ 3 \ 4; 5 \ 6 \ 7]$
- $b = [7 \ 5 \ 3; 8 \ 1 \ 2]$

9. What is the output of the following operation? $2 * a$ ($c = 2 * a$)

- If the values of the Matrix:-
- $a = [2 \ 3 \ 4; 5 \ 6 \ 7]$

10. What is the output of the following operation? $A * B$ ($C = A * B$)

- If the values of the Matrices are:-
- $A = [1 \ 4; 2 \ 0]$
- $B = [2 \ 1; 3 \ 4]$

11. What are the outputs of the following?

- $3i + 7 + 4i + 8$
- $3j + 9i + 3j + 7 + 8i$

12. What are the outputs of the following?

- $3i * 5i$
- $7j * 6i$
- $4i * 8j$
- $3 * 7i$
- $71j * 10$
- $3 * 9$

13. Evaluate the following

- $\log_{10}(10)$

- $\log_{10}(2)$
- $\log_{10}(-3)$
- $\log_{10}(2) * \log_{10}(2)$

14. What are the outputs of the following?

- $-7/i$
- $-8/4i$
- $-6j/3i$
- $-8/64j$

15. Evaluate the following through Octave

- $[2, 4, 5, 7] - [1, 0, 1, 3]$
- $[4, 3, 6, 3] + [2, 0, 1, 7]$
- $[2, 3, 5] - [0, 0, 1]$
- $[4, 3, 2] + [1, 1, 1]$

16. Evaluate the following through Octave

- $[2i, 4i, 3] * 6i$
- $[3, 9, 8j] / 4j$
- $[-4, -5i, -8] / 4i$
- $[-9, -6, -3] * 2j$

17. What is the difference between $\log(2)$ and $\log_{10}(2)$? Find the answer using Octave.

15. Evaluate the following expression in Octave

$$(3^2 + 7*5 - 9/3) / (4^2 - 3*(8 - 7/2) + 6)$$

Modify the equation to prioritize subtraction over addition. Rewrite it using parentheses to alter the precedence of operations and run it in octave

16. Evaluate the following using Octave:

$$((7^3 - 5^2 * (3 + 2^4)) / (4 + 3^3)) ^ (1/3) + \log_{10}(100) * \sin(\pi/6)$$

17. Using matrices:

$$A = [2, 4, 6; 8, 10, 12; 14, 16, 18];$$

$$B = [1, 3, 5; 7, 9, 11; 13, 15, 17];$$

Perform the following:

1. $A+B$
2. $A-B$
3. $A*B$

18. Given matrices:

$A = [2, 4, 6; 8, 10, 12; 14, 16, 18];$

$B = [1, 3, 5; 7, 9, 11; 13, 15, 17];$

Compute the element-wise addition, subtraction, and multiplication of A and B.

19. Evaluate using octave CLI

- $12^2 - 8*(15 - 4) + 3^4$
- $5^3 + (25/5) * (9 - 3^2)$
- $(7 + 4) * (3^2 - 5) / 2 + 10$

20. Evaluate using octave CLI

- $\cos(\pi/4)$
- $\sin(\pi/3) + \tan(\pi/6)$
- $2 * \cos(\pi/3) - 3 * \sin(\pi/6)$

21. Evaluate using octave CLI

- Create and display a 2x3 matrix
- Add two matrices
 - $A = [3, 5; 7, 9];$
 - $B = [1, 2; 4, 8];$
- Subtract matrices:
 - $X = [9, 8; 6, 4];$
 - $Y = [3, 2; 1, 0];$
- Multiply matrix $[2, 3; 4, 6]$ by 5

22. Evaluate

- $[1, 2, 3] .* [4, 5, 6]$
- $[12, 24, 36] ./ [3, 6, 9]$
- $(5 + 3i) + (2 - 4i)$
- $(9 + 7i) - (3 + 2i)$
- $(2 + 3i) * (4 - 5i)$
- $\log_{10}(100)$
- $\log(\exp(1))$
- $\log(20) * \log_{10}(10)$
- $[10, 20, 30] + [5, 15, 25]$
- $[50, 60, 70] - [10, 20, 30]$
- $4 * [1, 2, 3]$
- Multiply complex vector $[2i, 3i, 4]$ by 2
- Divide complex vector $[6i, 9i, 12]$ by 3
- Add real vector $[1, 2, 3]$ and complex vector $[2i, 3i, 4i]$